

Test Exercise 5

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$$\Pr[\text{resp}_i = 1] = \frac{\exp[\beta_0 + \beta_1 \text{male}_i + \beta_2 \text{active}_i + \beta_3 \text{age}_i + \beta_4 (\frac{\text{age}_i}{10})^2]}{1 + \exp[\beta_0 + \beta_1 \text{male}_i + \beta_2 \text{active}_i + \beta_3 \text{age}_i + \beta_4 (\frac{\text{age}_i}{10})^2]}$$

a) Let $\Pr[\text{resp}_i = 1] = p_i$

Then, $\Pr[\text{resp}_i = 0] = 1 - \Pr[\text{resp}_i = 1] = 1 - p_i$, since response is binary

Now let's take the marginal effects:

$$\frac{\partial \Pr[\text{resp}_i = 1]}{\partial \text{age}_i} + \frac{\partial \Pr[\text{resp}_i = 0]}{\partial \text{age}_i} = \frac{\partial p_i}{\partial \text{age}_i} + \frac{\partial (1 - p_i)}{\partial \text{age}_i} = \frac{\partial p_i}{\partial \text{age}_i} - \frac{\partial p_i}{\partial \text{age}_i} = 0$$

b) From (a), $\Pr[\text{resp}_i = 1] = p_i$ and $\Pr[\text{resp}_i = 0] = 1 - p_i$

Odds ratio: $\frac{\Pr[\text{resp}_i = 1]}{\Pr[\text{resp}_i = 0]} = \exp(x_i' \beta) \Leftrightarrow \frac{p_i}{1 - p_i} = \exp(x_i' \beta)$

$$\Leftrightarrow \boxed{\log \left[\frac{p_i}{1 - p_i} \right] = x_i' \beta} \quad \textcircled{1}$$

Now let's see what happens when $\text{resp}_i^{\text{new}} = 1 - \text{resp}_i$.

Odds ratio:
$$\frac{\Pr[\text{resp}_i^{\text{new}} = 1]}{\Pr[\text{resp}_i^{\text{new}} = 0]} = \exp(x_i' \beta_{\text{new}})$$

But now $\text{resp}_i^{\text{new}} = 1$ forces $\text{resp}_i = 0$

Similarly, $\text{resp}_i^{\text{new}} = 0$ forces $\text{resp}_i = 1$

Substituting,
$$\frac{\Pr[\text{resp}_i = 0]}{\Pr[\text{resp}_i = 1]} = \exp(x_i' \beta_{\text{new}}) \text{ or } \frac{1-p_i}{p_i} = \exp(x_i' \beta_{\text{new}})$$

$$\Rightarrow \boxed{\log\left[\frac{1-p_i}{p_i}\right] = x_i' \beta_{\text{new}}} \quad (2)$$

Now let's take
$$\frac{(1)}{(2)} \Rightarrow \frac{\beta}{\beta_{\text{new}}} = \frac{\log\left[\frac{p_i}{1-p_i}\right]}{\log\left[\frac{1-p_i}{p_i}\right]} = \frac{\log\left[\frac{p_i}{1-p_i}\right]}{\log\left[\left(\frac{p_i}{1-p_i}\right)^{-1}\right]} = \frac{\log\left[\frac{p_i}{1-p_i}\right]}{-\log\left[\frac{p_i}{1-p_i}\right]} = -1$$

So $\beta_{\text{new}} = -\beta$, which implies that the signs of all parameters change, while their absolute values remain the same.

c) In order to allow the age that maximizes the odds ratio to differ by gender, we could extend our logit specification by adding an interaction term between the male_i dummy and the age_i variable. A possible extension could be the following:

$$\frac{\Pr[\text{resp}_i=1]}{\Pr[\text{resp}_i=0]} = \exp \left[\beta_0 + \beta_1 \text{male}_i + \beta_2 \text{active}_i + \beta_3 \text{age}_i + \beta_4 \left(\frac{\text{age}_i}{10} \right)^2 + \beta_5 \text{male}_i \text{age}_i + \beta_6 \text{male}_i \left(\frac{\text{age}_i}{10} \right)^2 \right]$$